

The Making of Vehicle Currencies

Evidence from DeFi Route-Level Markets

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2026

Motivation: vehicle currencies are routing infrastructure

- A vehicle currency is not just a large asset or a popular endpoint.
- It is an asset that other trades route through when direct exchange is missing, thin, or costly.
- Classic foreign-exchange markets make the vehicle role hard to observe directly.
- DeFi gives transaction-level routes, liquidity, endpoint tokens, and settlement traces.

Research question

When and why does an asset become the intermediary through which other assets trade?

One-sentence thesis

Main claim

Vehicle currencies are made by route feasibility and liquidity concentration: they protect traders from missing or thin direct markets, persist with vehicle-linked liquidity, rotate under stress, and become more virtual when settlement architecture nets physical token movement.

- WETH and stablecoins are vehicle-currency candidates in the data.
- V3 and V4 are architecture changes that reveal how market design affects vehicle use and settlement.

Why DeFi is a useful laboratory

What traditional data often hide

- route choice
- intermediate asset identity
- direct-route availability
- execution-cost counterfactuals
- settlement implementation

What the DeFi data reveal

- reconstructed multi-hop routes
- exact source, destination, vehicle token
- pool liquidity and LP allocation
- route-cost quotes and no-direct cases
- receipt-level transfer incidence

Measurement: separate endpoint demand from bridge use

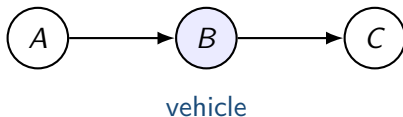
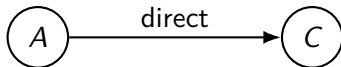
Main vehicle-use proxy

$$\text{BridgeShare}_{k,t} = \frac{\text{USD volume of indirect routes where } k \text{ is intermediate}}{\text{USD volume of all indirect routes}}$$

- Main object is conditional on indirect routing.
- All-route bridge share reports economic scope.
- Endpoint share is a control/diagnostic, not the vehicle proxy.

Authors' calculations from reconstructed Ethereum DEX route data.

Route objects



A token is counted as a vehicle only if it is an intermediate, not the source or destination.

Empirical spine

1. **Formation:** direct-market incompleteness and thin-direct protection.
Evidence: route availability, depth, and cost advantage predict vehicle use.
2. **Liquidity:** liquidity and route use persist together.
Evidence: LP concentration predicts BridgeShare; lagged BridgeShare predicts future LP.
3. **Stress:** risky incumbent loses share on impact.
Evidence: WETH-minus-stable share falls on stress days.
4. **Architecture:** direct-market deepening reduces vehicle dependence.
Evidence: post-V3 no-direct/WETH cases fall.
5. **Settlement:** netting separates route use from physical movement.
Evidence: V4 lowers intermediary-token transfer incidence.

Proposition 1: direct-market incompleteness

Statement

For an endpoint pair, a candidate vehicle expands the feasible execution set when the direct route is unavailable or thin. Conditional on both routes being feasible, the vehicle route improves execution only when its route cost is below the direct-route cost.

Empirical margins

- no-direct, vehicle-available cases
- thin direct route, vehicle route available
- common-support cost advantage
- vehicle-route depth

Result 1: formation loads on route opportunity

	Actual vehicle share	Vehicle share $t + 7$	Vehicle share $t + 30$
Route-cost advantage, per 100 bp	0.1462*** ($p < 0.001$)	0.0001*** ($p = 0.002$)	0.0002*** ($p < 0.001$)
Vehicle-route availability	19.1385*** ($p < 0.001$)	0.0940*** ($p < 0.001$)	0.1254*** ($p < 0.001$)
Vehicle-route depth	28.8502*** ($p < 0.001$)		
LP concentration		0.1117*** ($p < 0.001$)	
Lagged vehicle share			0.3630*** ($p < 0.001$)
FE	pair $\times$ date	token, date	token, date
Observations	189,143	9,380	9,265

Authors' calculations from reconstructed route-cost and liquidity data.

Result 1 interpretation

- The vehicle role is not simply “the cheapest route everywhere.”
- The strongest interpretation is **availability and thin-direct protection**.
- Common-support cost advantage is one mechanism, but not the whole vehicle value.
- The route-cost evidence therefore separates feasibility, thin-direct protection, and common-support price improvement.

Proposition 2: liquidity-route persistence

Statement

Vehicle-linked liquidity and vehicle use are jointly persistent: liquidity around a candidate vehicle predicts future vehicle use, and current vehicle use predicts future vehicle-linked liquidity.

- The empirical object is persistence in liquidity and route use.
- The estimates are predictive evidence rather than a liquidity-shock design.

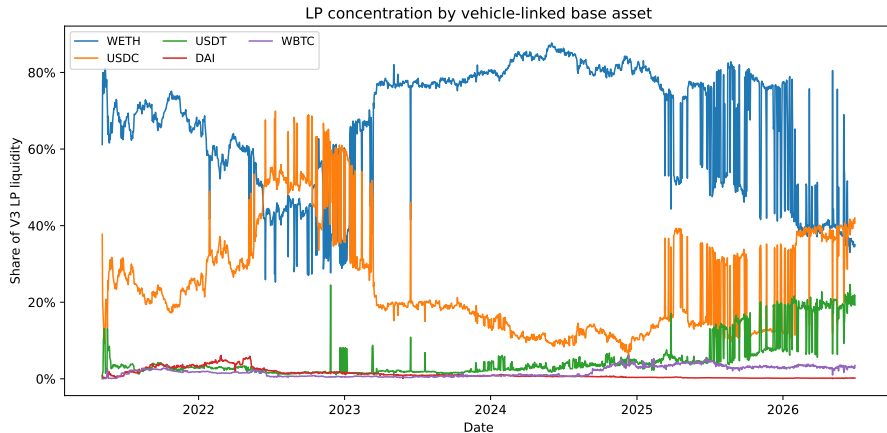
Result 2: liquidity and route use predict each other

	Vehicle share $t + 7$	LP concentration $t + 7$	Log LP liquidity $t + 7$
LP concentration	0.1117*** ($p < 0.001$)		
Lagged vehicle share		0.0419*** ($p < 0.001$)	1.1713*** ($p < 0.001$)
FE	token, date	token, date	token, date
SE clustering	date	date	date
Observations	9,380	9,380	9,380

- Liquidity concentration predicts future vehicle use.
- Vehicle use predicts future vehicle-linked liquidity.

Authors' calculations from reconstructed route and liquidity data.

Figure: vehicle shares and liquidity



Authors' calculations from reconstructed route and liquidity data.

Proposition 3: impact stress rotation

Statement

A risk or credibility shock to an incumbent vehicle reduces its route-intermediation share on impact relative to safer substitutes, conditional on common route opportunities.

- Model is interval-neutral: it predicts an impact response.
- Empirically strongest at the same-day event window.
- Hourly/weekly results are useful duration/noise diagnostics, not the headline.

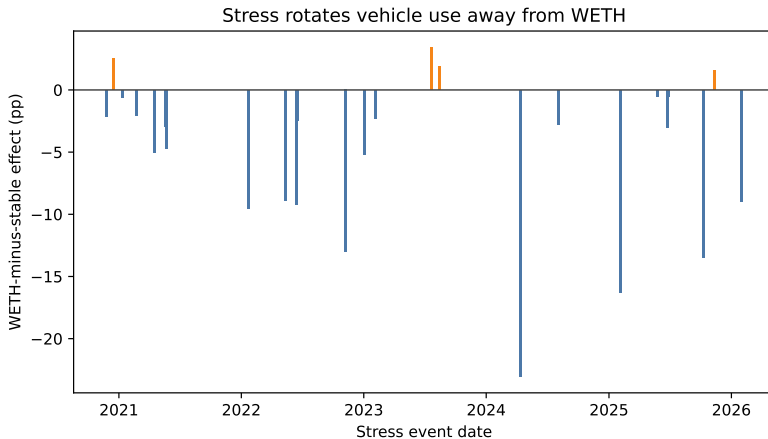
Result 3: stress rotates away from WETH

	WETH share	Stable share	WETH-stable gap	Direct share	Indirect volume
Stress event day	−1.48** pp ($p = 0.018$)	1.48** pp ($p = 0.018$)	−2.96** pp ($p = 0.018$)	−0.25 pp ($p = 0.679$)	0.51*** log pts ($p = 0.005$)

- Same-day common-support rotation is the cleanest stress result.
- Threshold and non-overlap checks remain negative and statistically significant.

Authors' calculations from common-support stress event design.

Figure: common-support stress rotation



Authors' calculations from common-support stress event design.

Proposition 4a: direct-market-deepening architecture

Statement

An architecture that increases direct pairwise depth reduces reliance on vehicle routes that exist only because the direct route is absent or thin.

- Empirical test bed: concentrated-liquidity launch.
- Main empirical claim: no-direct/WETH-available cases fall.
- The V3 evidence is strongest for direct-route opportunity, not for every architecture outcome.

Result 4a: V3 reduces no-direct dependence

	No-direct WETH	Q1 direct	Q1 no-direct	Q2 no-direct
Post V3	−25.81*** ($p < 0.001$)	56.42*** ($p < 0.001$)	−33.03*** ($p = 0.007$)	−47.51*** ($p < 0.001$)
FE	pair	pair	pair	pair
SE clustering	pair	pair	pair	pair

- Strongest message: architecture can reduce vehicle dependence by expanding direct-route opportunity.

Authors' calculations from pair-level route opportunity panel.

Proposition 4b: settlement netting

Statement

Settlement netting can preserve the route-pricing role of a vehicle while reducing physical transfers of the intermediary token.

- Empirical test bed: matched V3 and V4 route units.
- Route vehicle use and physical settlement movement become separable objects.

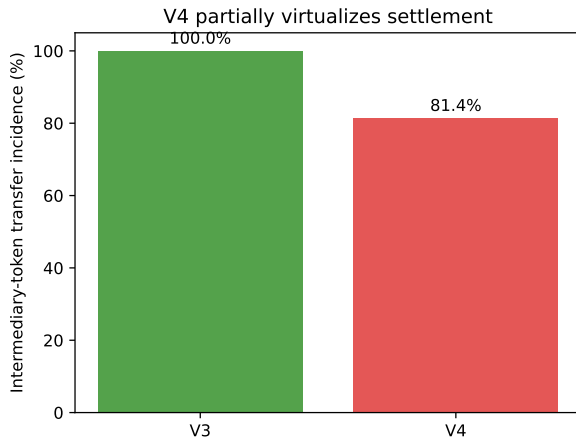
Result 4b: V4 virtualizes part of the vehicle role

	All sizes	Small	Medium	Large
V4 - V3 transfer incidence	-18.6*** pp ($p < 0.001$)	-39.4*** pp ($p < 0.001$)	-12.3*** pp ($p = 0.002$)	-3.2* pp ($p = 0.083$)
V3 transfer incidence	100.0%	100.0%	100.0%	100.0%
V4 transfer incidence	81.4%	60.6%	87.7%	96.8%

- V4 does not eliminate vehicle currencies.
- It makes part of the vehicle role virtual: the token can remain the route intermediary without moving physically in every route unit.

Authors' calculations from matched V3/V4 route units and transaction receipts.

Figure: settlement-transfer incidence



Authors' calculations from matched V3/V4 route units and transaction receipts.

Additional mechanism: common vehicle-linked liquidity

	Full	Stress	High veh.	Low veh.	Excl. top
Market factor	0.7063*** ($p < 0.001$)	0.6987*** ($p < 0.001$)	0.6572*** ($p < 0.001$)	0.5276*** ($p < 0.001$)	0.7062*** ($p < 0.001$)
Vehicle factor	0.1626*** ($p < 0.001$)	0.1586*** ($p < 0.001$)	0.2174*** ($p < 0.001$)	-0.0921 ($p = 0.307$)	0.1670*** ($p < 0.001$)
Vehicle factor $\times$ stress		0.1064** ($p = 0.030$)			

- Same-vehicle pools share a liquidity component beyond market-wide liquidity.
- The vehicle-linked component strengthens where vehicle use is high and during stress.

Authors' calculations from reconstructed vehicle-linked liquidity panel.

What the evidence supports

1. **Formation:** route opportunity and depth predict vehicle use.
2. **Persistence:** vehicle use and vehicle-linked liquidity are persistent together.
3. **Rotation:** stress moves flow away from risky incumbent vehicle use toward stable substitutes on impact.
4. **Architecture:** market design changes direct-route feasibility.
5. **Settlement:** netting separates route intermediation from physical token movement.

Scope of the evidence

- Route-cost estimates are scoped to quoteable venues; venue coverage is reported separately.
- Liquidity estimates measure persistence and predictability rather than shock-based causal effects.
- Stress rotation is an impact-window result, with hourly and weekly diagnostics used to study duration.
- V4 evidence measures intermediary-token transfer incidence and matched route use while separating route intermediation from physical settlement.
- Curve and Fluid are included in realized vehicle-use measurement and excluded from executable-depth quote panels.

Contribution

- **Vehicle-currency literature:** measures vehicle use directly as route intermediation.
- **Liquidity/intermediation literature:** shows vehicle status is tied to liquidity concentration and route opportunity.
- **Market design:** shows architecture changes vehicle reliance and settlement implementation.
- **DeFi evidence:** turns transparent route-level ledgers into a laboratory for classic monetary intermediation questions.

Conclusion

Takeaway

Vehicle currencies are not just dominant assets. They are route-level liquidity institutions whose role depends on direct-market incompleteness, liquidity concentration, stress-state substitution, and settlement architecture.